Market Rate for Services

Rates below are in **SGD** and reflect the Singapore market (SGT). All figures are indicative benchmarks; actual rates depend on scope complexity, timeline, and engagement model (project-based vs. retainer).

1. Full Stack Development

Scale & Scope

Scale	Duration	Description
Small	2–6 weeks	Landing pages, internal dashboards (single module), simple CRUD APIs, static sites with CMS backends.
Mid	2–6 months	Multi-tenant SaaS modules, admin portals, real-time dashboards, API gateways with auth, mobile-first PWAs.
Large	6–18 months	Enterprise platforms (multi-service, role-based access), marketplace/multi-vendor systems, end-to-end digital transformation.

Typical Cost by Stack (Mid-Scale Reference)

Stack	Technologies	Indicative Cost (SGD)	Notes
React + Node/Express + PostgreSQL	TypeScript, REST/GraphQL, Prisma/Knex	$50K–$150K	Most common startup stack; fast iteration.
React + Python (FastAPI/Django) + PostgreSQL	TypeScript, Python, Celery, Redis	$60K–$180K	Better for data-heavy or ML-adjacent apps.
React + .NET (C#) + SQL Server	ASP.NET Core, Entity Framework, Azure	$80K–$220K	Enterprise/ gov-facing; higher licensing & hosting overhead.
React Native / Flutter + Backend	Cross-platform mobile + any backend	$70K–$200K	Add ~30–50% premium over web-only.
Electron + React + Node	Desktop app (Windows/macOS/Linux)	$80K–$250K	Niche skill set; higher testing & distribution complexity.

Rate Models

- **Hourly:** $80–$150/hr (senior full stack, Singapore)
- **Monthly retainer (full-time equivalent):** $12K–$22K/mo
- **Fixed-price:** Add 20–30% contingency over time-and-materials estimate

2. Disruption Routing

Specialised public-transport disruption intelligence service combining **RAPTOR-based point-to-point routing** with **LTA DataMall real-time disruption detection**.

Core Capabilities

Capability	Description
RAPTOR Routing Engine	Multi-criteria transit routing (time, transfers, walking distance). Supports GTFS-static schedules with dynamic recalculation on disruption events.
LTA DataMall Integration	Real-time bus arrival, train service alerts, carpark availability, traffic images. Polling + webhook-style event processing pipeline.
Disruption Detection	Anomaly detection on ETA drift, missing trips, service status API flags. Configurable thresholds with false-positive suppression.
Alternative Routing	Automatic re-routing on detected disruption. Suggests alternative stations, bus bridging, or mode-shift (MRT ↔ bus).

Indicative Cost

Engagement	Scope	Cost (SGD)
Proof of Concept	Single transit line, RAPTOR prototype, DataMall polling	$25K–$45K
Production System	Full Singapore network (MRT + LRT + bus), real-time disruption pipeline, web/mobile UI	$80K–$180K
Ongoing Maintenance	GTFS schedule updates, API changes, model retuning	$3K–$6K/mo

Hourly Rate

- $100–$180/hr (specialised domain: transit algorithms + real-time data engineering)

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3. AI Feature Integration

End-to-end integration of AI/LLM capabilities into existing products — from architecture to production operations.

Skills & Coverage

Skill Area	What It Covers
Harness Engineering	Multi-agent orchestration, sub-agent delegation, tool-use frameworks, streaming SSE pipelines, agent memory (SQLite/vector DB). Design of parent/child agent architectures with tool access control.
Model Selection	Evaluate and benchmark LLMs across cost/latency/accuracy trade-offs. Experience with DeepSeek, GPT-4o, Claude, Gemini, open-source models (Llama, Qwen). On-prem vs. cloud deployment analysis.
Prompt Engineering	System prompt design for constrained tool-use agents. Few-shot structuring, chain-of-thought reasoning patterns, structured output formatting (JSON mode, function-calling schemas). Iterative prompt evaluation and regression testing.

Skill Area	What It Covers
RAG & Embeddings	Retrieval-augmented generation pipelines, chunking strategies, hybrid search (semantic + keyword), embedding model selection and fine-tuning.
AI Safety & Guardrails	Output validation, content filtering, rate limiting, token budget enforcement, human-in-the-loop approval workflows.
MLOps / LLMOps	Model versioning, A/B evaluation, cost monitoring and token usage tracking, automated regression suites for prompt changes.
Backend Integration	Streaming endpoints (SSE/WebSocket), tool execution sandboxing, state machine-based agent loops, structured logging for agent audit trails.

Indicative Cost

Engagement	Scope	Cost (SGD)
AI Readiness Assessment	Audit existing product, recommend AI integration points, architecture proposal	$8K–$18K
Feature Prototype	Single AI feature end-to-end (e.g., "AI code review assistant", "chat-with-your-data")	$25K–$60K
Full Integration	Multi-agent system, RAG pipeline, production deployment with monitoring	$80K–$200K

Hourly Rate

- $120–$200/hr (covers architecture, prompt engineering, agent design, and productionization)

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4. Bug Hunting & Fixing

Systematic debugging of production issues, performance bottlenecks, and edge-case failures.

Approach

- **Triage & Reproduce** — isolate root cause with minimal reproduction steps.
- **Scientific Debugging** — hypothesis → instrumentation → evidence collection → fix → regression test.
- **Tooling** — runtime log injection (debug server), browser console/network trace analysis, Electron main+renderer debugging, SQL query profiling, React component re-render analysis.

Typical Bugs

Type	Examples
Logic / State	Stale closures, race conditions, incorrect state transitions, React re-render loops
Concurrency	SSE stream desync, WebSocket reconnect storms, terminal buffer corruption
Performance	Memory leaks (Node/Electron), slow SQL queries (missing indices, N+1), large DOM trees

Type	Examples
Integration	API schema mismatches, auth token expiry not handled, CORS/network errors
Platform	Electron IPC failures, native module compilation, OS-specific path/encoding bugs

Rate Models

- **Hourly (ad-hoc):** $100–$160/hr
- **Bug Bounty (per fix):** $500–$5,000 depending on severity (P0 critical → P3 cosmetic)
- **Retainer (dedicated debugging bandwidth):** $6K–$12K/mo for 40–80 hrs

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5. Small Neural Network Training & Inference

Training and deploying compact neural models for specialised, low-latency use cases.

Scope

Area	Examples
Tabular / Structured Data	ETA prediction, demand forecasting, fraud/anomaly detection on time-series
NLP (compact)	Text classification, entity extraction, intent recognition for domain-specific chatbots
Reinforcement Learning	Small-scale decision optimisation (e.g., routing policy learning)
On-Device Inference	Model quantisation, ONNX/TensorFlow Lite export, edge deployment for latency-sensitive apps

Indicative Cost

Engagement	Scope	Cost (SGD)
Feasibility Study	Data audit, baseline model, feasibility report	$8K–$18K
Model Development	Data prep, architecture selection, training, hyperparameter tuning, evaluation	$20K–$60K
Production Deployment	Inference API, model serving, monitoring (drift detection), retraining pipeline	$15K–$40K

Hourly Rate

- $100–$180/hr

Rate Summary Table

Service	Hourly (SGD)	Monthly Retainer	Project Range
Full Stack Development	$80–$150	$12K–$22K	$20K–$250K

Service	Hourly (SGD)	Monthly Retainer	Project Range
Disruption Routing	$100–$180	$3K–$6K (maintenance)	$25K–$180K
AI Feature Integration	$120–$200	Negotiable	$8K–$200K
Bug Hunting & Fixing	$100–$160	$6K–$12K	$500–$5K (per fix)
Small Neural Network	$100–$180	Negotiable	$8K–$100K

Notes: All rates in Singapore Dollars (SGD). Prices are indicative and assume senior-level delivery. Rush/premium timelines typically add 25–50%. Government/GLC clients may have additional compliance and procurement lead-time considerations.